Spike Train Analysis

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Spike rates and intervals

Waveforms and cell types

Connectivity and Cross correlograms

Population coding and embedding

Decoding

Conclusion

Position of the problem

- Preprocessing is done (spike sorting, histology, behavior,...)
- We have unit spike trains and other timestamps stored in some format...
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Position of the problem

- Preprocessing is done (spike sorting, histology, behavior,...)
- We have unit spike trains and other timestamps stored in some format...
- We have a research question.



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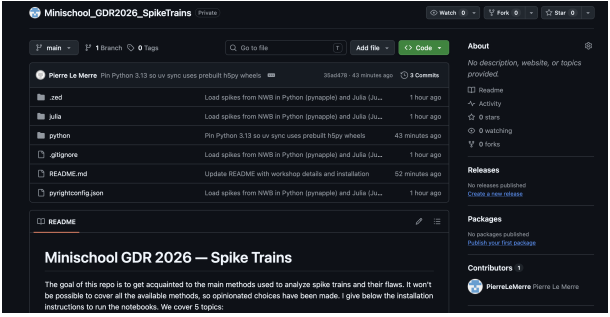
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Go to ⇒
`https://github.com/samuelgarcia/school_neuralnet_paris_2026`
folder `notebooks_spiketrain_analysis`



Download the data - Proposed framework



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Buccino et al, 2022

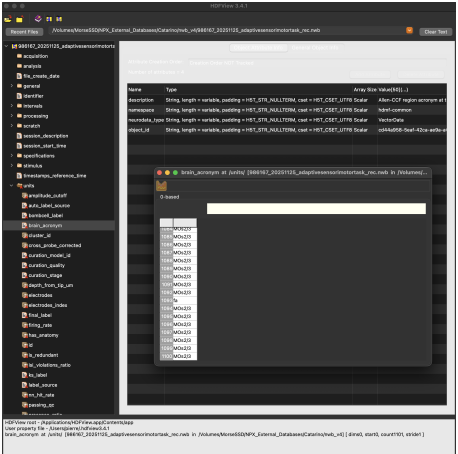
https://spikeinterface.readthedocs.io/en/stable/ https://nwb.org/ https://dandiarchive.org/

Download the data

link 2 data

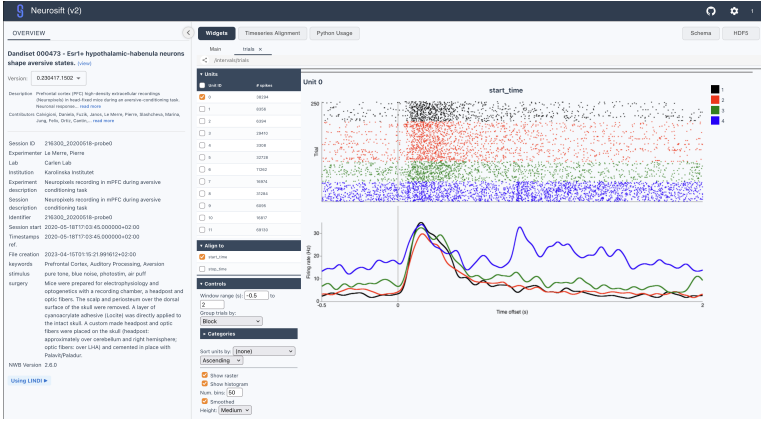
Explore NWB data - HDFView

https://github.com/HDFGroup/hdfview/releases/tag/v3.5.0

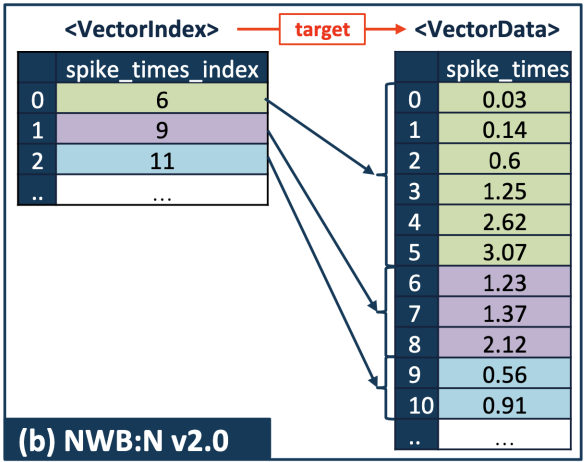


Explore NWB data - Neurosift

https://neurosift.app/



Explore NWB data - where are my spike times?



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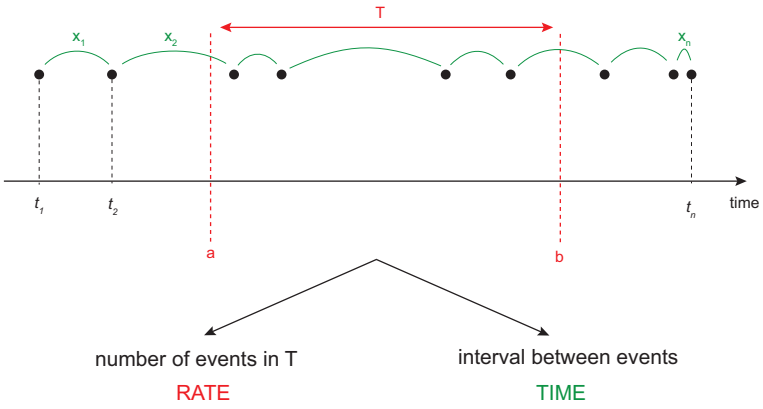
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Spike trains are Point processes



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